



# Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

## ACADEMIC— UNDER GRADUATE STUDIES DIVISION

---

### First Semester 2026-27

### Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

**Course Number** : EEE U111

**Course Title** : ELECTRICAL SCIENCES

**Instructor-in-charge** : ABHIJIT ASATI ([abhijit\\_asati@pilani.bits-pilani.ac.in](mailto:abhijit_asati@pilani.bits-pilani.ac.in), 2220-Q)

L1: Abhijit Asati ([abhijit\\_asati@pilani.bits-pilani.ac.in](mailto:abhijit_asati@pilani.bits-pilani.ac.in), 2220-Q),

L2: Puneet Mishra ([puneet.mishra@pilani.bits-pilani.ac.in](mailto:puneet.mishra@pilani.bits-pilani.ac.in), 2210-B )



# Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

## ACADEMIC— UNDER GRADUATE STUDIES DIVISION

### [Team of Tutorial Instructors]

:

#### Tutorial:

- T1: Puneet Mishra ([puneet.mishra@pilani.bits-pilani.ac.in](mailto:puneet.mishra@pilani.bits-pilani.ac.in), 2210-B )  
T2: Satyendra K Mourya ([satyendra.mourya@pilani.bits-pilani.ac.in](mailto:satyendra.mourya@pilani.bits-pilani.ac.in), 2210-D)  
T3: Bhavesh Chauhan ([bhavesh.chauhan@pilani.bits-pilani.ac.in](mailto:bhavesh.chauhan@pilani.bits-pilani.ac.in), 2220-G),  
T4: Dheerendra Singh ([dhs@pilani.bits-pilani.ac.in](mailto:dhs@pilani.bits-pilani.ac.in), 2220-O)  
T5: Ashish Patel ([ashish.patel@dhs@pilani.bits-pilani.ac.in](mailto:ashish.patel@dhs@pilani.bits-pilani.ac.in), 2210-O)  
T6: Pawan K Ajmera ([pawan.ajmera@pilani.bits-pilani.ac.in](mailto:pawan.ajmera@pilani.bits-pilani.ac.in), 2136)  
T7: Hari Om Bansal ([hbansal@pilani.bits-pilani.ac.in](mailto:hbansal@pilani.bits-pilani.ac.in), 1209-B)  
T8: Bhavesh Chauhan ([bhavesh.chauhan@pilani.bits-pilani.ac.in](mailto:bhavesh.chauhan@pilani.bits-pilani.ac.in), 2220-G)  
T9: Pawan K Ajmera ([pawan.ajmera@pilani.bits-pilani.ac.in](mailto:pawan.ajmera@pilani.bits-pilani.ac.in), 2136)  
T10: Govind Prasad ([govind.prasad@pilani.bits-pilani.ac.in](mailto:govind.prasad@pilani.bits-pilani.ac.in), 2223-A)  
T11: Aditya R Gautam ([aditya.gautam@pilani.bits-pilani.ac.in](mailto:aditya.gautam@pilani.bits-pilani.ac.in), 2210-R)

#### Laboratory:

- P1: UDAY MITTAL (OLAB:[P20240046@pilani.bits-pilani.ac.in](mailto:P20240046@pilani.bits-pilani.ac.in), RKIC-IOT LAB)  
Amit (ITECH:[P20250022@pilani.bits-pilani.ac.in](mailto:P20250022@pilani.bits-pilani.ac.in), 2212)  
P2 Manish Prasad (OLAB:[p20240903@pilani.bits-pilani.ac.in](mailto:p20240903@pilani.bits-pilani.ac.in), RKIC-IOT LAB)  
Shreya Chandravanshi ITECH:([p20230440@pilani.bits-pilani.ac.in](mailto:p20230440@pilani.bits-pilani.ac.in), 2145)  
P3: PRAVEEN KUMAR (ITECH:[P20250804@pilani.bits-pilani.ac.in](mailto:P20250804@pilani.bits-pilani.ac.in), 2212)  
Himanshu Shekhar (OLAB:[p20250084@pilani.bits-pilani.ac.in](mailto:p20250084@pilani.bits-pilani.ac.in), RKIC-IOT LAB)  
P4: DIVYA VYAS (OLAB:[p20210431@pilani.bits-pilani.ac.in](mailto:p20210431@pilani.bits-pilani.ac.in), 2212)  
POOJA MAURYA (ITECH:[p20210429@pilani.bits-pilani.ac.in](mailto:p20210429@pilani.bits-pilani.ac.in), 2213)  
P5: ANTONY DAVID (ITECH:[p20240439@pilani.bits-pilani.ac.in](mailto:p20240439@pilani.bits-pilani.ac.in), 2212)  
Himanshu Shekhar (OLAB:[p20250084@pilani.bits-pilani.ac.in](mailto:p20250084@pilani.bits-pilani.ac.in), RKIC-IOT LAB)  
P6: Virender Singh (OLAB:[P20240442@pilani.bits-pilani.ac.in](mailto:P20240442@pilani.bits-pilani.ac.in), 2212)  
ANITA GAHLOT (ITECH:[p20240448@pilani.bits-pilani.ac.in](mailto:p20240448@pilani.bits-pilani.ac.in), 2212)  
P7: Virender Singh (OLAB:[P20240442@pilani.bits-pilani.ac.in](mailto:P20240442@pilani.bits-pilani.ac.in), 2212)  
ANITA GAHLOT (ITECH:[p20240448@pilani.bits-pilani.ac.in](mailto:p20240448@pilani.bits-pilani.ac.in), 2212)  
P8: LAXMI LAXMI ( OLAB:[p20240043@pilani.bits-pilani.ac.in](mailto:p20240043@pilani.bits-pilani.ac.in), 2211)  
HARSHITA SAROJ (ITECH:[p20240443@pilani.bits-pilani.ac.in](mailto:p20240443@pilani.bits-pilani.ac.in), 3129)  
P9: Rohit Kumar (OLAB:[p20220401@pilani.bits-pilani.ac.in](mailto:p20220401@pilani.bits-pilani.ac.in), 2145)  
HARSHITA SAROJ (ITECH:[p20240443@pilani.bits-pilani.ac.in](mailto:p20240443@pilani.bits-pilani.ac.in), 3129)  
P10: PRAVEEN KUMAR (ITECH:[P20250804@pilani.bits-pilani.ac.in](mailto:P20250804@pilani.bits-pilani.ac.in), 2212)  
UDAY MITTAL (OLAB:[P20240046@pilani.bits-pilani.ac.in](mailto:P20240046@pilani.bits-pilani.ac.in), RKIC-IOT LAB)  
P11: Amit (ITECH:[P20250022@pilani.bits-pilani.ac.in](mailto:P20250022@pilani.bits-pilani.ac.in), 2212)  
Rohit Kumar Shaw (OLAB:[P20250908@pilani.bits-pilani.ac.in](mailto:P20250908@pilani.bits-pilani.ac.in), 2145)

- 1. Course description:** Course covers basic passive circuit elements, dependent and independent sources, network theorems, circuit analysis techniques and transient response of first and second order circuits. Sinusoidal steady-state analysis and AC power analysis. Semiconductors - operation of diodes (ideal and real), Zener diodes and its applications. Introduction of BJTs, Introduction to operational amplifiers and applications.



# Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

## ACADEMIC— UNDER GRADUATE STUDIES DIVISION

**2. Scope and Objective:** The primary objective of this course is to teach the principles of different aspects of electrical sciences: (1) Circuits Analysis both steady state and transient (2) Electronic Devices: Diode, BJT and OPAMP.

**3. Course learning outcomes:**

1. Students will learn the circuit analysis (voltage, current and power calculation) techniques for DC and AC (1-phase) circuits using Kirchhoff's current law (KCL), Kirchhoff's voltage, law (KVL), Ohms law, Nodal analysis, Mesh analysis, Network theorems (Thevenin's theorem, Norton's theorem, Maximum power transfer theorem, Superposition theorem) etc.
2. Students will be able to perform circuit analysis on electronic circuit consisting of elements such as diode, BJT transistor, OPAMP.

**4. Text books and references:**

**TB1.** Leonard S. Bobrow and Navneet Gupta, Foundations of Electrical Engineering, Oxford University Press, Asian Edition, 2015

**RB1.** Alexander and Sadiku, Fundamentals of Electric Circuits, Indian Edition, Mc Graw Hill, 2019

**RB2.** Allan R Hambley, Electrical Engineering: Principles and Applications; 7th Edition (global edition), Pearson education limited, 2019.

**5. Course Format & Weekly Workflow**

Course work involves 2 lecture hour of class work, 1 tutorial hour (engagement in problem solving) and 2 Laboratory hour (engagement in laboratory experiments).

**6. Course Plan:**

| Week No. | Module                                     | Lec. | Coverage                                                                                                                                                         | Ref. (TB) | Learning Outcome                                                                                                                                                       |
|----------|--------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-2      | Basic electrical circuit laws and elements | 1-2  | Voltage and current sources, basic circuit components, Kirchhoff's current law (KCL), Kirchhoff's voltage, law (KVL), Instantaneous power, Inductors, Capacitors | 1.1-1.7   | Students will learn basic circuit elements and the laws of electrical science to solve basic electrical circuits which also includes the concepts of dependent sources |



# Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

## ACADEMIC— UNDER GRADUATE STUDIES DIVISION

|      |                                      |       |                                                                                                                                       |                        |                                                                                                                                         |
|------|--------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
|      |                                      | 3     | Independent and dependent sources                                                                                                     | 1.8                    | Students will learn independent and dependent sources, source transformation.                                                           |
| 2-4  | Principles of circuit analysis       | 4-5   | Nodal analysis, Mesh analysis,                                                                                                        | 2.1-2.3                | Students will be able to solve circuits by nodal and mesh analysis.                                                                     |
|      |                                      | 6-8   | Network theorems (Thevenin's theorem, Norton's theorem, Source transformation, Maximum power transfer theorem, Superposition theorem) | 2.4,2.6                | Students will be able to solve circuits by nodal and mesh analysis and will be able to understand the applications of network theorems. |
| 5-6  | Time domain analysis                 | 9-12  | Transient response of first and second order circuits (natural and forced response)                                                   | 3.2-3.5                | Students will be able to perform time domain analysis of first and second order circuits.                                               |
| 6-8  | AC analysis                          | 13-14 | Time-domain analysis, Phasor representation of alternating quantities, $j$ operator and Phasor algebra, Frequency-domain analysis     | 4.1-4.3                | Students will learn concept of phasor, phasor algebra, frequency domain analysis of AC circuits.                                        |
|      |                                      | 15-17 | AC powers: Average power, apparent power and complex power, Power factor corrections                                                  | 4.4-4.5                | Students will learn concepts of power in AC circuits.                                                                                   |
| 9-11 | Semiconductor: Diodes and Transistor | 18-20 | Basics of PN junction, Junction diode, Ideal diode and Applications (rectifiers and clippers)                                         | 6.2, (partly) 6.3, 6.4 | Students will study basics of semiconductors, diodes and their use in various electronic circuits.                                      |
|      |                                      | 21-22 | Zener diode (its model and applications in                                                                                            | 6.6                    | Students will learn applications of Zener                                                                                               |



# Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

## ACADEMIC— UNDER GRADUATE STUDIES DIVISION

|    |                                               |       |                                                      |         |                                                                         |
|----|-----------------------------------------------|-------|------------------------------------------------------|---------|-------------------------------------------------------------------------|
|    |                                               |       | voltage regulator and clipper)                       |         | diodes in electronic circuits.                                          |
|    |                                               | 23-24 | BJTs (circuits for different modes of operation)     | 7.1-7.3 | Students will learn the purpose of different modes of operation in BJT. |
| 12 | Operational amplifier and Digital Electronics | 25-27 | Basics of operational amplifier and its application, | 10.1    | Students will learn about op-amp circuits and applications.             |

### 7. Laboratory Work:

Students will carry out 10 hardware and software experiments from the following topics: KCL/KVL verification, Dependent and independent sources, Nodal and mesh analysis, Network theorems (Thevenin's, Norton, Superposition, Max. power transfer), Transient and AC analysis, Diode circuits, BJT circuits and Operational amplifiers circuits. A soft copy of manual will be provided to students for preparing and maintaining the laboratory records. Take printouts of these manuals bind it in advance for maintaining the lab record. On the day of the experiment, note practical reading in a copy and also do its theoretical analysis in the same copy, then, get it fair in the lab manual at home/hostel and in the next lab session get it evaluated by the Lab instructor.

### 8. Evaluation Scheme:

| Sr. No. | Evaluation Component         | Duration | Weightage | Marks (300)    | Date & Time                   | Nature of Component |
|---------|------------------------------|----------|-----------|----------------|-------------------------------|---------------------|
| 1.      | Class participation*         | *        | 5 %       | 15             | During lecture Hour           | -                   |
| 2.      | Tutorial Quizzes             | 15 Min.  | 15 %      | 45 (15xBest 3) | During Tutorial Hour          | Closed Book         |
| 3.      | Lab Regularity               | Lab hour | 20 %      | 60 (=6x10)     | During Lab Hour               | Open Book           |
| 4.      | Lab Comprehensive Evaluation | 2 Hour   | 10 %      | 30             | During Lab Comprehensive Exam | Closed Book         |
| 3.      | Mid-Sem. Exam                | 90 Min.  | 20 %      | 60             | 9/10/2026 (AN2)               | Open Book           |
| 4.      | Comprehensive Exam.          | 180 Min. | 30 %      | 90             | 14/12/2026 (AN)               | Closed Book         |



# Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

## ACADEMIC— UNDER GRADUATE STUDIES DIVISION

\* Class Participation: This component evaluates student engagement during lecture hours. It will include short quizzes or discussions, conducted throughout the semester during lecture hour. Active and consistent participation will contribute to this component of the overall assessment.

**Criterion for NC:** A student should obtain 30 % of the average of the top three performers or 25 % of the course total marks, whichever is lower to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

9. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
10. **Chamber consultation hours:** Displayed on Nalanda (<https://nalanda.bits-pilani.ac.in>)
11. **Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for mid semester or comprehensive exam only.
12. **Notices:** Displayed on Nalanda.

*Abhijit Asati*

Instructor-in-charge

EEE U111